ECON 2050 – Mathematical Economics

Linear Algebra II

In the previous lecture, we discussed problems of the following kind:

> Given an $m \times n$ matrix $\boldsymbol{A}$ and $\boldsymbol{b} \in \mathbb{R}^m$, find an $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{Ax} = \boldsymbol{b}$ or prove that no such $\boldsymbol{x}$ exists.

Recall that $\boldsymbol{Ax} = \boldsymbol{b}$ represents a system of linear equations, and therefore these problems are ultimately about whether a system of linear equations has a solution or not.

Detour: Comparison of vectors

Recall that if $\boldsymbol{x} \in \mathbb{R}^n$ is a vector, then the j-th element/coordinate/entry of $\boldsymbol{x}$ is denoted x_j, where $j = 1, \ldots, n$.

For two vectors $\boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$, where $n \geq 2$, we define

- $\boldsymbol{y} = \boldsymbol{z}$ if $y_i = z_i$ for all $i = 1, \ldots, n$,
- $\boldsymbol{y} \geq \boldsymbol{z}$ if $y_i \geq z_i$ for all $i = 1, \ldots, n$,
- $\boldsymbol{y} > \boldsymbol{z}$ if $y_i \geq z_i$ for all $i = 1, \ldots, n$ and $y_j > z_j$ for *at least one* j, and
- $\boldsymbol{y} \gg \boldsymbol{z}$ if $y_i > z_i$ for all $i = 1, \ldots, n$.

Recall also that $\mathbb{R}^n_+ \equiv \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{x} \geq \boldsymbol{0}\}$.

Non-negative solutions

In economics, finance, and business, oftentimes it doesn't make sense for the relevant quantities to take negative values.

- For example, quantities of consumption, capital, labor, assets, to name a few.

Therefore, we sometimes wish to understand whether we can find a **non-negative** solution to a system of linear equations, or no solution exists.

- More formally, we wish to find a $\boldsymbol{x} \in \mathbb{R}^n$ with $\boldsymbol{x} \geq \boldsymbol{0}$ such that $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ or show that no such $\boldsymbol{x}$ exists.
- Observe that if $\boldsymbol{b} = \boldsymbol{0}$, we can just let $\boldsymbol{x} = \boldsymbol{0}$, so we assume that $\boldsymbol{b} \neq \boldsymbol{0}$ henceforth.
- The problem can be framed as follows: can $\boldsymbol{b}$ be expressed as a *non-negative* linear combination of the columns of $\boldsymbol{A}$; that is, do there exist $x_1, x_2, \ldots, x_n \geq 0$ such that

$$\boldsymbol{b} = x_1\boldsymbol{a}^1 + x_2\boldsymbol{a}^2 + \ldots + x_n\boldsymbol{a}^n.$$

To that end, it is useful to define the following:

Definition

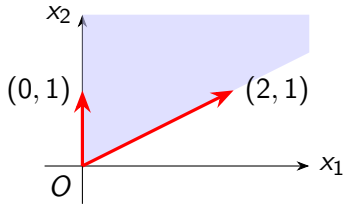
The set of all *non-negative* linear combinations of the columns of $\boldsymbol{A}$ is called the **cone** generated by the columns of $\boldsymbol{A}$. It is denoted cone($\boldsymbol{A}$).

Example: Consider the matrix $\boldsymbol{A} = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$. The cone generated by the columns of $\boldsymbol{A}$ is

$$\text{cone}(\boldsymbol{A}) = \{a_1(0,1) + a_2(2,1) : a_1, a_2 \in \mathbb{R}_+\} = \{(2a_2, a_1 + a_2) : a_1, a_2 \in \mathbb{R}_+\}.$$

- Because $a_1, a_2 \geq 0$, every vector in cone($\boldsymbol{A}$) must be in the non-negative orthant (also called the first quadrant).
- Also because $a_1, a_2 \geq 0$, for every vector $\boldsymbol{x} = (x_1, x_2) \in \text{cone}(\boldsymbol{A})$, we must have $2x_2 = 2(a_1 + a_2) \geq 2a_2 = x_1$.

Example (cont'd): Therefore, the cone generated by the columns of $\boldsymbol{A}$ is the shaded region in the picture below:



In words, cone($\boldsymbol{A}$) is the area between the rays from the origin passing through $(0,1)$ and $(2,1)$, respectively.

It is instructive to compare the definition of span($\boldsymbol{A}$) with cone($\boldsymbol{A}$) for an $m \times n$ matrix $\boldsymbol{A}$:

$$\text{span}(\boldsymbol{A}) = \{\boldsymbol{y} \in \mathbb{R}^m : \boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} \text{ for some } \boldsymbol{x} \in \mathbb{R}^n\},$$
$$\text{cone}(\boldsymbol{A}) = \left\{\boldsymbol{y} \in \mathbb{R}^m : \boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} \text{ for some } \boldsymbol{x} \in \mathbb{R}^n_+\right\}.$$

To see the difference, let's consider the following two matrices:

$$\begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

- The span of the columns of the left matrix is the entire $\mathbb{R}^2$, while the cone generated by its columns is the shaded area in the previous slide.
- The span of the columns of the right matrix is the entire $\mathbb{R}^2$, while the cone generated by its columns is the non-negative orthant (also called the first quadrant).

Farkas lemma

Theorem (Farkas Lemma)

Let $\boldsymbol{A}$ be an $m \times n$ matrix, $\boldsymbol{b} \in \mathbb{R}^m$, and $F = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{Ax} = \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\}$. Then either $F \neq \varnothing$ or there exists $\boldsymbol{y} \in \mathbb{R}^m$ such that $\boldsymbol{yA} \geq \boldsymbol{0}$ and $\boldsymbol{y} \cdot \boldsymbol{b} < 0$ but not both.

It is useful to rephrase Farkas lemma in the following way:

> Either the system of linear equations $\boldsymbol{Ax} = \boldsymbol{b}$ has a *non-negative* solution, or the system $\boldsymbol{yA} \geq \boldsymbol{0}$, $\boldsymbol{y} \cdot \boldsymbol{b} < 0$ has a solution, *but not both*.

The system $\boldsymbol{yA} \geq \boldsymbol{0}$ and $\boldsymbol{y} \cdot \boldsymbol{b} < 0$ is sometimes referred to as the **Farkas alternative**.

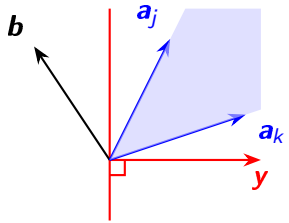
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(Algebraic) intuition:

- Take any linear combination of the equations in $\boldsymbol{Ax} = \boldsymbol{b}$ to get $\boldsymbol{yAx} = \boldsymbol{y} \cdot \boldsymbol{b}$.
- Suppose the first system has non-negative solution, then it must also be a solution to the second.
- If we can choose y so that $\boldsymbol{yA} \geq \boldsymbol{0}$ and $\boldsymbol{y} \cdot \boldsymbol{b} < 0$, we find that the left-hand side of the single equation $\boldsymbol{yAx} = \boldsymbol{y} \cdot \boldsymbol{b}$ is at least zero while the right-hand side is strictly negative, a contradiction.
- Thus, the first system cannot have a non-negative solution.

Geometric interpretation:



- The fact that the system $\boldsymbol{Ax} = \boldsymbol{b}$ has a non-negative solution can be interpreted as $\boldsymbol{b} \in \text{cone}(\boldsymbol{A})$.
- Therefore, $F = \varnothing$ implies that $\boldsymbol{b} \notin \text{cone}(\boldsymbol{A})$.
- In this case, the Farkas lemma asserts that there is a vector $\boldsymbol{y}$ that makes a non-obtuse angle with every column of $\boldsymbol{A}$ (and hence $\boldsymbol{yA} \geq \boldsymbol{0}$) and an obtuse angle with $\boldsymbol{b}$ (and hence $\boldsymbol{y} \cdot \boldsymbol{b} < 0$).

Example: We use the Farkas lemma to decide the solvability of the system:

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \\ 1 \end{pmatrix}.$$

We are interested in non-negative solutions of this system. The Farkas alternative is

$$\begin{pmatrix} y_1 & y_2 & y_3 & y_4 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \text{and} \quad 2y_1 + 2y_2 + 2y_3 + y_4 < 0.$$

As a system of inequalities the alternative is (ignore the strict inequality for the moment):

$$\begin{cases} y_1 + y_2 + y_4 \geq 0 \\ y_1 + y_3 + y_4 \geq 0 \\ y_2 + y_3 + y_4 \geq 0 \end{cases} \Rightarrow \begin{cases} y_1 + y_2 \geq -y_4 \\ y_1 + y_3 \geq -y_4 \\ y_2 + y_3 \geq -y_4 \end{cases}$$

To maximize the chance of satisfying the strict inequality $2y_1 + 2y_2 + 2y_3 + y_4 < 0$, we let y_1, y_2, y_3 to be "as negative as they can" (i.e., without violating the other three inequalities):

- To that end, we let all three inequalities on the previous slide hold with equality; adding them together yields $2y_1 + 2y_2 + 2y_3 = -3y_4$.
- Consequently, it must be that $2y_1 + 2y_2 + 2y_3 + y_4 < 0$, implying that the alternative must have a solution. In particular, one solution is $y_1 = y_2 = y_3 = -1$, and $y_4 = 2$.

By the Farkas lemma, the original system has no solution.

In a vague sense, the solution to the alternative we identify provides an "explanation" for why the (original) system has no solution:

- Multiply the first three equations by -1 and add them together to yield $-2x_1 - 2x_2 - 2x_3 = -6$.
- However, this is inconsistent with the fourth equation multiplied by 2: $2x_1 + 2x_2 + 2x_3 = 2$.

Exercise: Use the Farkas lemma to decide if the following system has a non-negative solution:

$$\begin{pmatrix} 4 & 1 & -2 \\ 1 & 0 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}.$$

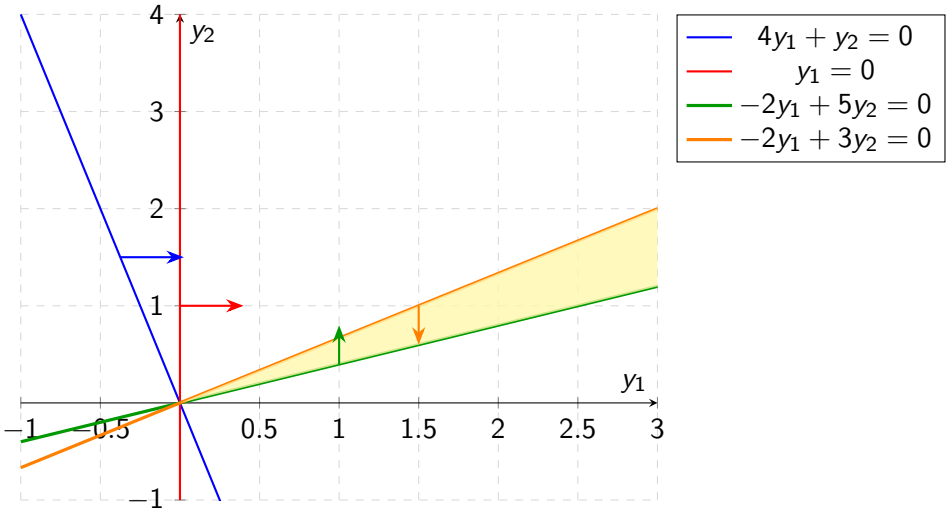
The Farkas alternative is

$$\begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{pmatrix} 4 & 1 & -2 \\ 1 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}, \quad \text{and} - 2y_1 + 3y_2 < 0.$$

As a system of inequalities the alternative is:

$$\begin{aligned} 4y_1 + y_2 &\geq 0, \\ y_1 + 0y_2 &\geq 0, \\ -2y_1 + 5y_2 &\geq 0, \\ -2y_1 + 3y_2 &< 0. \end{aligned}$$

By graphing the inequalities on a plane (see next slide), we see that the alternative has a solution. By the Farkas lemma, the original system has no solution.


$4y_1 + y_2 = 0$
$y_1 = 0$
$-2y_1 + 5y_2 = 0$
$-2y_1 + 3y_2 = 0$
y_2
y_1
4
3
2
1
−1
−1
−0.5
0.5
1
1.5
2
2.5
3

Linear inequalities

Sometimes we also wish to understand the solvability of systems of linear inequalities. E.g.,

> Given an $m \times n$ matrix $\boldsymbol{A}$ and $\boldsymbol{b} \in \mathbb{R}^m$, find an $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$ or prove that no such $\boldsymbol{x}$ exists.

In what follows we show that the version of Farkas lemma we discussed above (for equality systems) can also be used to derive the appropriate alternatives for inequality systems.

Below is a cookbook procedure that allows us to convert any inequality system into an equality system with non-negative variables:

- For each inequality of the system, $\sum_{j=1}^{n} a_{ij}x_j \leq b_i$, we can add a **slack variable** to it to get $\sum_{j=1}^{n} a_{ij}x_j + s_i = b_i$.
- For each inequality of the system of the form $\sum_{j=1}^{n} a_{ij}x_j \geq b_i$, we can subtract a **surplus variable** to it to get $\sum_{j=1}^{n} a_{ij}x_j - s_i = b_i$.
- For a variable, x_j, that is unrestricted in sign can be replaced by two non-negative variables z_j and z_j' by setting $x_j = z_j - z_j'$.

Example: Let us derive the Farkas alternative for the system $[\boldsymbol{Ax} \leq \boldsymbol{b}$ and $\boldsymbol{x} \geq \boldsymbol{0}]$.

- Recall that we can add a slack variable $s_1 \geq 0$ to the first inequality $\sum_{j=1}^{n} a_{1j}x_j \leq b_1$ to convert it to an equality $\sum_{j=1}^{n} a_{1j}x_j + s_1 = b_1$.
- We do this to all inequalities, and let $\boldsymbol{s} \equiv (s_1, s_2, \ldots, s_m)$ be the *slack vector*, evidently, $\boldsymbol{s} \geq \boldsymbol{0}$.
- Therefore, "whether we can find $\boldsymbol{x} \geq \boldsymbol{0}$ such that $\boldsymbol{Ax} \leq \boldsymbol{b}$" is equivalent to "whether we can find $\boldsymbol{x} \geq \boldsymbol{0}$ and $\boldsymbol{s} \geq \boldsymbol{0}$ such that $\boldsymbol{Ax} + \boldsymbol{s} = \boldsymbol{b}$."
- Let $\boldsymbol{I}$ denote the $(m \times m)$ identity matrix, which is a matrix whose diagonal entries are 1 and off-diagonal entries are 0. It is not hard to verify that for any $\boldsymbol{c} \in \mathbb{R}^m$, $\boldsymbol{Ic} = \boldsymbol{cI} = \boldsymbol{c}$. Thus, $\boldsymbol{Ax} + \boldsymbol{s} = \boldsymbol{b}$ is equivalent to $\boldsymbol{Ax} + \boldsymbol{Is} = \boldsymbol{b}$.
- Set $\boldsymbol{B} \equiv [\boldsymbol{A}|\boldsymbol{I}]$, and let $\boldsymbol{z} \equiv (\boldsymbol{x}, \boldsymbol{s}) = (x_1, x_2, \ldots, x_n, s_1, s_2, \ldots, s_m)$, and we can write the system $[\boldsymbol{Ax} + \boldsymbol{Is} = \boldsymbol{b}$, $\boldsymbol{x} \geq \boldsymbol{0}$, and $\boldsymbol{s} \geq \boldsymbol{0}]$ as $[\boldsymbol{Bz} = \boldsymbol{b}$ and $\boldsymbol{z} \geq \boldsymbol{0}]$.

Example (cont'd):

- Now apply the Farkas lemma to the system $[\boldsymbol{Bz} = \boldsymbol{b} \text{ and } \boldsymbol{z} \geq \boldsymbol{0}]$ to get the Farkas alternative, namely the system $[\boldsymbol{yB} \geq \boldsymbol{0} \text{ and } \boldsymbol{y} \cdot \boldsymbol{b} < 0]$.
- Since $\boldsymbol{B} = [\boldsymbol{A}|\boldsymbol{I}]$, $\boldsymbol{y}[\boldsymbol{A}|\boldsymbol{I}] = \boldsymbol{yB} \geq \boldsymbol{0}$, implying that $\boldsymbol{yA} \geq \boldsymbol{0}$ and $\boldsymbol{yI} = \boldsymbol{y} \geq \boldsymbol{0}$.
- Thus, the Farkas alternative for the system $[\boldsymbol{Ax} \leq \boldsymbol{b} \text{ and } \boldsymbol{x} \geq \boldsymbol{0}]$ is

$$[\boldsymbol{yA} \geq \boldsymbol{0}, \quad \boldsymbol{y} \geq \boldsymbol{0}, \quad \text{and} \quad \boldsymbol{y} \cdot \boldsymbol{b} < 0].$$

The punchline here is that by a judicious use of auxiliary variables one can convert almost anything into the system of equations we started with, namely $[\boldsymbol{Ax} = \boldsymbol{b} \text{ and } \boldsymbol{x} \geq \boldsymbol{0}]$, and then apply the Farkas lemma.

Application: Arbitrage

Arbitrage is the practice of taking advantage of a difference in prices in two or more markets.

- In the long run, arbitrage opportunities shouldn't prevail: investors take advantages of such opportunities until the prices are adjusted and they disappear.
- In other words, arbitrage opportunities can only exist for an extremely brief period of time: Almost as soon as they are discovered, they are wiped out as a result of the response of investors.
- The absence of arbitrage opportunities is the driving principle of financial theory.

In economics and finance, we say that an arbitrage opportunity exists if there exists an investment portfolio that pays off non-negative amounts in every scenario that has a strictly negative cost.

To formally model arbitrage, suppose there are m possible assets whose payoff depends on a scenario that realizes in the future, and there are n possible scenarios.

- Let a_{ij} be the payoff from one share of asset i in scenario j; the **payoff matrix $\boldsymbol{A}$** summarizes all possible payoffs:

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

- A portfolio of assets is represented by a vector $\boldsymbol{y} \in \mathbb{R}^m$ where the i-th component, y_i, represents the amount of asset i held.
- $y_i > 0$ represents buying asset i, and $y_i > 0$ represents selling it.

"No arbitrage" requires that there doesn't exist a portfolio with a strictly negative cost that pays off non-negative amounts in every scenario.

If $\boldsymbol{p} > 0$ is a vector of asset prices, we can state the no arbitrage condition algebraically as follows:

$$\boldsymbol{yA} \geq \boldsymbol{0} \Rightarrow \boldsymbol{y} \cdot \boldsymbol{p} \geq 0$$

Equivalently, the system $[\boldsymbol{yA} \geq \boldsymbol{0}$ and $\boldsymbol{y} \cdot \boldsymbol{p} < 0]$ must have no solution.

From the Farkas lemma, there must exist a non-negative vector $\boldsymbol{\pi} \in \mathbb{R}^n_+$ such that $\boldsymbol{p} = \boldsymbol{A\pi}$.

- Scaling if necessary, $\boldsymbol{\pi}$ can be interpreted as a *probability vector* (i.e., $\sum_{j=1}^{m} \pi_i = 1$) that specifies the chance of each scenario realizes.
- That is, π_j is the chance (or probability) of the scenario j for each $j = 1, \ldots, n$.
- Therefore, this condition specifies that there exist chances of all scenarios (or, more formally, a probability distribution over scenarios) under which every asset's expected value is equal to its price (in other words, every investment has an expected return of zero).
- An investor who using these chances would conclude that the assets are *fairly priced*.

The discussion above gives rise to the following famous result in finance.

Theorem (Arbitrage Theorem)

Let $\boldsymbol{A}$ be an $m \times n$ payoff matrix, and let $\boldsymbol{p}$ denote the vector of (relative) asset prices. Exactly one of the following statements holds:

(a) *There exists an investment portfolio $\boldsymbol{y} \in \mathbb{R}^m$ such that $\boldsymbol{yA} \geq \boldsymbol{0}$ and $\boldsymbol{y} \cdot \boldsymbol{p} < 0$.*

(b) *There is a probability vector $\boldsymbol{\pi} \in \mathbb{R}^n_+$ (with $\sum_{j=1}^n \pi_j = 1$) such that $\boldsymbol{p} = \boldsymbol{A\pi}$.*

Intuitively, the theorem states that

- either the expected value of all possible assets is the same and equal to their prices,
- or it is possible to create an arbitrage portfolio at a strictly negative cost that guarantees a non-negative return for every possible scenario,

but not both. In other words, no arbitrage opportunities implies that market is "fair" in the sense that all assets are fairly priced.

Pricing (call) options: The Black–Scholes formula

Perhaps the most remarkable application of the no arbitrage idea is to the pricing of derivative securities called options.

- A **call option** on an asset is a contract giving one the "option" to buy the asset at a specified price, called the **strike price**, at a specified time in the future.
- The advantage of a call option is that it allows one to postpone the purchase of an asset until after one sees the price.
- In particular, one can wait for the market price to rise above the strike price, and then exercise the option. That is, buy at the strike price and resell at the higher market price.

Question: How much is a call option worth?

For simplicity, assume a single time period and a market consisting of a stock, a bond, and a call option on the stock.

Let K be the strike price of the call option and suppose that it can be exercised only at the end of the time period. Suppose S is the value of the stock at the end of the time period.

- If $S > K$, the option holder will exercise the call option yielding a profit of $S - K$ dollars.
- If $S \leq K$, the option holder will let the call option expire and its value to the holder is zero.
- To summarize, the call option has a payoff of $\max\{0, S - K\}$.

Suppose that there are only two possible future scenarios, say good and bad.

- The good scenario is where the stock goes up by a factor $u > 1$, and the bad scenario is where the stock declines by a factor $d < 1$.
- Investing in the bond is risk free, which means that in all cases in the future the value of the bond goes up by a factor $r > 1$; we assume that $r < u$ to make the setting interesting.
- Let S_0 be the initial stock price and B the price of the bond.

We now have an economy with three assets (stock, bond, call option) and two possible scenarios; that is, $m = 3$ and $n = 2$. The payoff matrix $\boldsymbol{A}$ can be written as

$$\begin{pmatrix} uS_0 & dS_0 \\ rB & rB \\ \max\{0, uS_0 - K\} & \max\{0, dS_0 - K\} \end{pmatrix}.$$

If $\boldsymbol{p} = (p_1, p_2, p_3)$ is the vector of prices then $p_1 = S_0, p_2 = B$ and p_3 is the price of the call option we wish to determine.

Since it's reasonable to assume that there is no arbitrage opportunities in the market, the arbitrage theorem implies the existence of non-negative π_1, π_2 such that

$$\begin{pmatrix} uS_0 & dS_0 \\ rB & rB \\ \max\{0, uS_0 - K\} & \max\{0, dS_0 - K\} \end{pmatrix} \begin{pmatrix} \pi_1 \\ \pi_2 \end{pmatrix} = \begin{pmatrix} S_0 \\ B \\ p_3 \end{pmatrix}.$$

Consider the first two equations:

$$uS_0\pi_1 + dS_0\pi_2 = S_0$$
$$rB\pi_1 + rB\pi_2 = B$$

They have a unique non-negative solution

$$\pi_1 = \frac{r-d}{r(u-d)}, \quad \text{and} \quad \pi_2 = \frac{u-r}{r(u-d)}.$$

Using this solution, we deduce that

$$p_3 = \pi_1 \max\left\{0, uS^0 - K\right\} + \pi_2 \max\left\{0, dS^0 - K\right\}. \qquad (\star)$$

This option pricing formula $(\star)$ is the discrete (one period) analogue of the celebrated Black–Scholes formula.

- Note that the price of the call option doesn't depend on the probabilities of either the good or bad scenario being realized.